

Adityakeerti

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Professional Summary

Final-year AI/ML student focused on bridging the gap between complex AI models and production-ready software. Hands on experience in designing scalable backend architectures, high-throughput event processing, and robust data pipelines to support autonomous agents and computer vision applications. Passionate about building reliable, real-world systems that deliver measurable business impact.

Technical Skills

Languages: Python, Java, SQL, TypeScript, Dart

Backend & Architecture: FastAPI, Spring Boot, REST APIs, API Security, System Design

Databases: PostgreSQL, MySQL, SQLite

AI & Processing: PyTorch, OpenCV, YOLO, LLM Orchestration, HuggingFace

Cloud & DevOps: Docker, Google Cloud (Vertex AI), Git, GitHub Actions, CI/CD, Postman

Testing: PyTest, JUnit, Integration Testing, API Testing

Experience

Peerprep

Software Engineering Intern

Mar 2026 - May 2026

- Built an automated ingestion engine using Python to standardize and extract data from unpredictable, multi-format academic documents.
- Implemented strict validation checks, reducing per-record processing time from 10 minutes to ~30 seconds while maintaining 99.98% accuracy on internal validation datasets.

Tech Stack: Python, OpenCV, OCR, Git, GitHub

Projects

R.A.G.E. — Windows Automation Agent (AI Desktop Assistant) [GitHub](#)

May 2026 - Present

- Developed a fully autonomous Windows agent that executes complex, multi-step goals (e.g., compiling daily briefs, scraping web pages, batch-tiling apps) by translating open-ended user commands into sequential system actions.
- Implemented an adaptive ReAct (Reason+Act) feedback loop, enabling the agent to observe active window handles, read system errors, self-correct dynamically, and process natural language modifications on the fly.
- Built proactive background daemons for download organization, clipboard monitoring, and OS notification interception, backed by an SQLite transaction ledger for instant rollbacks and regex-based safety filters blocking known high-risk shell operations.

Tech Stack: Python, TypeScript, React, SQLite, Win32 API, pywebview, CustomTkinter

DOC-OC - Automated Marksheet Extraction System [GitHub](#) | [Live Demo](#)

Sep 2025 - Oct 2025

- Engineered a multi-board extraction pipeline utilizing OpenCV, custom YOLOv8 object detection, and Hugging Face Table Transformers to automate structured data parsing for administrative workflows.
- Built a custom structure-retaining OCR engine (via LLM Whisperer) integrated with strict fraud-mitigation checks to ensure data integrity across consecutive semesters.
- Developed a React/TypeScript and async FastAPI portal backed by a secure MySQL database, driving a 30x reduction in processing time (10 mins to ~30s per marksheet).

Tech Stack: Python, TypeScript, React, FastAPI, YOLOv8, HuggingFace, OpenCV, MySQL

Advanzia AutoLend - Autonomous Credit Limit Optimization [GitHub](#) | [Live Demo](#)

Jan 2026

- Implemented a Dueling Double DQN-based risk optimization engine inspired by existing research on dynamic credit limit optimization, combined with a Cox Proportional Hazard Model to estimate default probabilities.
- Architected a high-throughput microservices backend (Spring Boot, FastAPI) leveraging Kafka event streams and Redis caching, benchmarking throughput at 5,000+ requests/min with sub-300ms inference latency during evaluation.
- Designed an immutable double-entry ledger architecture with audit trails and built a simulation pipeline modeling 10,000+ synthetic users to pre-train and validate reinforcement learning policies.

Tech Stack: Java, Spring Boot, Python, FastAPI, PyTorch, Kafka, Redis, PostgreSQL

Education

Graphic Era Hill University

B.Tech Computer Science — Artificial Intelligence and Machine Learning

CGPA: 7.47/10.0

Dehradun

2023 – Present

Achievements

- 1st Place, MariTHON National Hackathon** — SOF extraction pipeline built and deployed using Google Cloud Vertex AI for the Integrated Maritime Exchange.
- Finalist, HackTheWinter National Hackathon** — AI-driven dynamic credit limit optimization system using reinforcement learning.
- Finalist, India Innovates 2026** — Qualified for the Physical Finale of the World's Largest Civic Tech Hackathon (Top 500 Teams) | Bharat Mandapam, New Delhi, India.